User Manual

Active Splitter

ASP-015-50-A



User Manual – Version v1.2.5
06 Nov 2026

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1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the Active Splitter. It forms an integral part of the product and must remain accessible to the personnel responsible for the device. If the device is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.

Before performing any operation, the entire manual must be read carefully and the relevant sections reviewed in detail.

1.1.1 User Qualification

The device is intended for use by qualified personnel familiar with laboratory electronic instrumentation and responsible for the installation, operation, and maintenance of the device throughout its service life. It is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the device only under supervision by qualified personnel.

1.1.2 Revision History

Document Revision: v1.2.5- 06 Nov 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period (10 years). During this period, documentation may be requested from the manufacturer.

1.2 Symbols and Conventions

The following graphical symbols and textual conventions may appear in this manual or on the device.

Provides important supplementary information.

Indicates procedures that, if not observed, may result in damage to the device or associated equipment.

Indicates procedures that, if not observed, may result in injury or serious damage.



(a) General hazard symbol



(b) Mandatory reading of the manual

Figure 1.1: The device may include the pictograms shown above.

All safety symbols must remain clearly visible throughout the lifetime of the device.

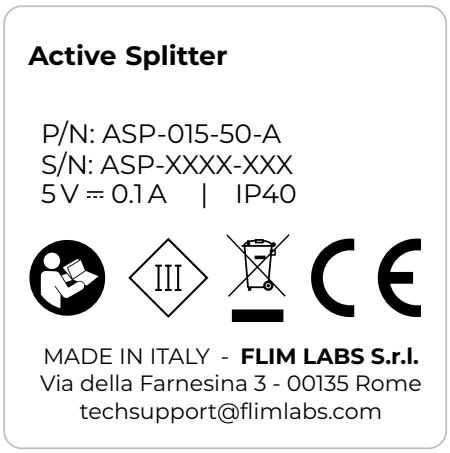


Figure 1.2: Product label

1.3 Manufacturer Information

Manufacturer:	FLIM LABS S.r.l.
Address:	Via della Farnesina, 3 00135 Rome (RM), Italy
Telephone:	+39-329 761540
E-mail:	info@flimlabs.com
Website:	www.flimlabs.com

1.4 Product Label

The product label affixed to the device contains the required regulatory markings and identification information including:

- Product name and model
- Manufacturer identification
- Regulatory markings
- Electrical specifications

An example product label is shown in Figure 1.2. Disposal requirements associated with the WEEE marking are described in Section 7.2.

Safety pictograms performing a protective function must not be removed, covered, or damaged. If the label becomes illegible or damaged, contact authorized service for replacement.

1.5 Declaration of Conformity

The device is supplied with a Declaration of Conformity drawn up in accordance with applicable European legislation. Before use, verify that the Declaration of Conformity is present. A copy of the Declaration of Conformity is shown in Figure 1.3.

1.6 Warranty and Liability

1.6.1 Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the device is operated and maintained according to this manual. Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty. Full warranty terms are available in the manufacturer's General Terms and Conditions published on the official website.

To submit a warranty claim, provide the following:

- Product type/version
- Proof of purchase (including date)
- Detailed description of the issue

EU Declaration of Conformity

CE

Device Information	
Type:	Active splitter
Model:	Active Splitter
Part Number:	ASP-015-50-A
Serial Number:	As indicated on the product label

Manufacturer Information	
Manufacturer:	FLIM LABS S.r.l.
Address:	Via della Farnesina, 3 00135 ROME (RM), Italy
Telephone:	+39 – 329 761540
E-mail:	info@flimlabs.com
Website:	www.flimlabs.com

The present declaration of conformity is issued under the sole responsibility of the manufacturer. The device complies with the following directives and regulations:

Applicable Directives and Regulations

- Directive 2014/35/EU, known as the “Low Voltage Directive”
- Directive 2014/30/EU, known as the “Electromagnetic Compatibility Directive”
- Directive 2011/65/EU, known as “RoHS”
- Delegated Directives (EU) 2015/863 and (EU) 2017/2102 amending Directive 2011/65/EU
- Directive 2012/19/EU, known as “WEEE”
- Regulation (EU) 2023/988 on General Product Safety

The manufacturer undertakes to provide, upon a duly reasoned request from the national authorities, all relevant information regarding the device.

Signed for and on behalf of FLIM LABS S.r.l.:

Place and Date	**Signature**
Rome, 24/10/2025	Alessandro Rossetta Alessandro Rossetta, CEO

Figure 1.3: EU Declaration of Conformity

- Operation outside specified electrical or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts
- Integration into systems not compliant with applicable safety regulations

The user is responsible for ensuring that the device is installed, operated, and integrated in accordance with this manual and all applicable local, national, and international regulations.

Except where otherwise required by mandatory law, the manufacturer shall not be liable for indirect, incidental, or consequential damages, including but not limited to loss of data, loss of profit, or system downtime.

2 Device Description

This section describes the functional characteristics, intended application, and technical features of the Active Splitter.

2.1 Functional Description

The Active Splitter is a compact electronic module designed for the controlled duplication and distribution of high-frequency electronic signals. It is primarily used in scientific instrumentation, timing systems, event counting, and fluorescence lifetime imaging microscopy (FLIM) applications, where reliable signal replication without degradation of timing or electrical characteristics is required.

The device is housed in a compact aluminum enclosure suitable for benchtop laboratory use or integration into portable measurement systems. The enclosure provides mechanical robustness and electromagnetic shielding within the specified operating limits. Power is supplied via USB Type-C at 5 V DC.

The Active Splitter features two bidirectional SMA connectors operating in loop-through configuration and one dedicated SMA output. The input signal can be applied to either of the loop-through ports and is forwarded unchanged to the second port. In parallel, the signal is actively replicated and provided at the dedicated output as an impedance-matched copy.

This architecture enables signal distribution to multiple instruments while maintaining signal integrity and proper impedance matching. The device is intended for immediate integration into measurement chains and does not require user configuration.

2.2 Intended Use

The Active Splitter is intended as a compact electronic module for the distribution and duplication of high-frequency signals by professional users in research laboratories, companies, and universities. It may be used as a stand-alone unit or integrated into custom measurement systems and instrumentation.

2.3 Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

2.4 Components Overview

Figure 2.1 shows the key components of the device, including:

1. USB Type-C power input
2. SMA output (active replicated signal)
3. SMA loop-through port 1 and 2
4. Mounting holes for mechanical fixing

2.5 Mechanical Drawings and Dimensions



Figure 2.1: Top view of the device showing the key components of the Active Splitter.

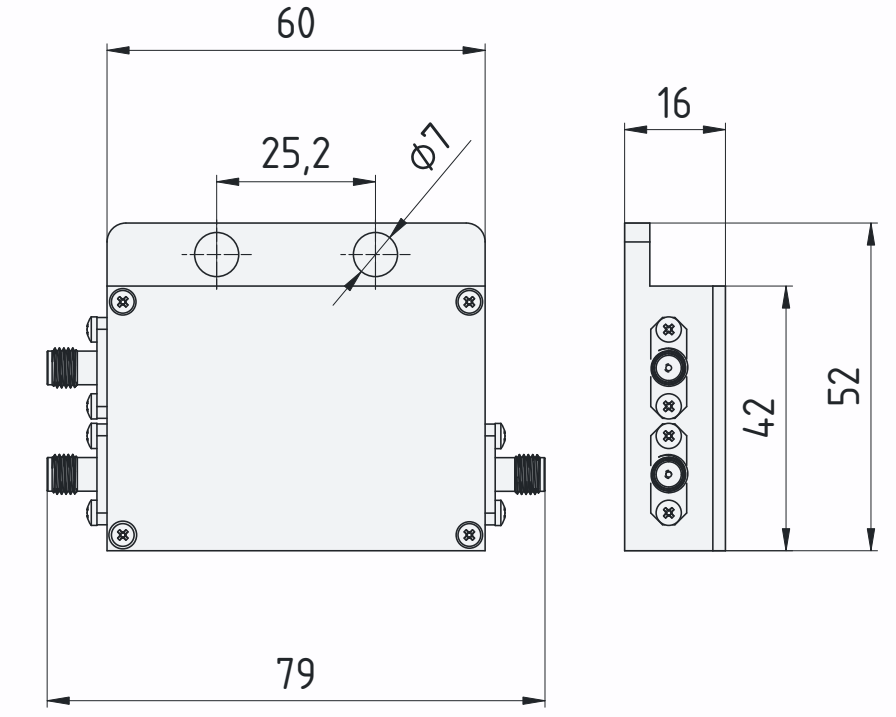


Figure 2.2: Technical drawing of the Active Splitter.

2.6 Technical Specifications

Item	ASP-015-50-A
Electrical Specifications	
Electrical safety class	Class III
Power supply	USB Type-C, 5 V DC
Current consumption (max)	0.1 A
Power consumption (max)	0.5 W
Signal Specifications	
Input connection	1x SMA
Input/Output connections	1x SMA (loop-through), 1x SMA output
Input polarity	positive or negative
Maximum input pulse height (abs)	5 V
Impedance	50 Ω
Timing variation	<5 ps
Lower frequency limit	30 kHz
Output loss P1dB	>0 dBm OP 1 dB
Maximum output amplitude	0.87 V rms
Physical Specifications	
Dimensions (L x W x H)	60 mm x 42 mm x 14 mm
Weight	80 g
Protection rating	IP40
Environmental Specifications	
Operating temperature	0 °C to +40°C
Storage temperature	-20 °C to +70°C
Humidity	up to 80% non-condensing

Table 2.1: Technical specifications of the Active Splitter.

3 Safety Information

3.1 General Safety Instructions

Before installation and operation, ensure that this manual is complete and legible. If any part of the documentation is missing or unclear, contact the manufacturer before proceeding.

The device must be used only as described in this manual. Any operation not explicitly described in this manual is considered improper use.

This manual complements applicable national and local safety regulations. If local regulations impose stricter requirements, those regulations take precedence.

Despite the implemented safety measures, residual risks may remain if the device is not used as specified.

3.2 Electrical Safety

- Operate the device only with the specified power supply.
- Use undamaged, properly rated cables.
- Disconnect power before modifying signal connections.
- Do not open the enclosure. There are no user-serviceable internal components.
- Keep away from water sources and other liquids.
- Unauthorized repair or modification is not permitted.

3.3 Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.

3.4 Modification and Improper Use

The device must not be modified, altered, or repaired by unauthorized personnel. Use outside the intended application, operation beyond specified electrical limits, or failure to follow the procedures described in this manual may result in malfunction and void the warranty.

3.5 Documentation Retention

This manual must be kept accessible to personnel responsible for installation, operation, and maintenance. A replacement copy may be requested from the manufacturer if required.

4 Setup

This section describes the correct procedure for unpacking, installing, and configuring the device before operation.

4.1 Scope of Delivery

- Active Splitter
- USB Type-C power cable
- Quick start guide
- Declaration of conformity

Verify that all items are present before proceeding with installation.

4.2 Unpacking and Inspection

The device is supplied in protective packaging to protect it during storage and transport. Handle the device carefully during unpacking to avoid mechanical damage.
After removing the packaging:

- Inspect the device and accessories for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that connectors and cables are intact.

If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.
Dispose of packaging materials responsibly and keep them out of reach of children.
The device is compact and lightweight and may be moved manually. When handling, support it securely to prevent accidental dropping.

4.3 Installation Requirements

Before installation and operation, ensure that the environmental conditions specified in Section 3.3 are met. Children must not have access to the installation or storage area. The device requires a regulated 5 V DC power source via USB Type-C.



Before connecting the device, verify that the input signal characteristics are compatible with the technical specifications provided in Section 2.6.

4.4 Mechanical Installation

Place the device on a stable laboratory surface with adequate space for cable routing and ventilation.
Ensure that:

- The device is positioned to prevent mechanical stress on connectors.
- Cables are routed safely and do not create tripping hazards.
- The power supply location is dry and away from water sources (e.g., sinks or splash zones).

Remove any protective dust or moisture covers before operation.

4.5 Electrical Connections

All electrical connections must be performed by qualified personnel who have read this manual. Connector locations are shown in Figure 2.1.
Proceed as follows:

1. **Power Connection**:
   a) Connect the USB Type-C cable to the device power input.

b) Connect the other end to a suitable 5 V DC power source such as a computer USB port, a power bank, or a certified 5 V DC adapter.

c) Verify visually that the LED power indicator confirms correct device power-up.

2. **Signal Connection**:

a) Connect the signal to one of the SMA loop-through ports.

b) Connect the second loop-through port as required in the signal chain.

c) Connect the dedicated SMA output to the measurement or processing device.

4.6 Example Connection Diagrams

Figure 4.1 shows a typical use case for the Active Splitter to enable simultaneous FLIM measurements and acquisition by the microscope system. The components shown in the diagram perform the following functions:

Acquisition of Photomultiplier Signals

PMT The photomultiplier tube converts light into measurable electrical signals for fluorescence imaging.

Active Splitter Duplicates the PMT signal to integrate the FLIM upgrade while maintaining parallel acquisition by the microscope system.

Wideband Amplifier Amplifies weak fluorescence signals with minimal added noise, supporting accurate lifetime measurements.

Constant Fraction Discriminator Converts amplified PMT signals into photon-counting outputs, enabling the FLIM DAQ to record precise timestamps of single fluorescence photons.

The example is provided for reference. The actual configuration may vary depending on the connected equipment and application.

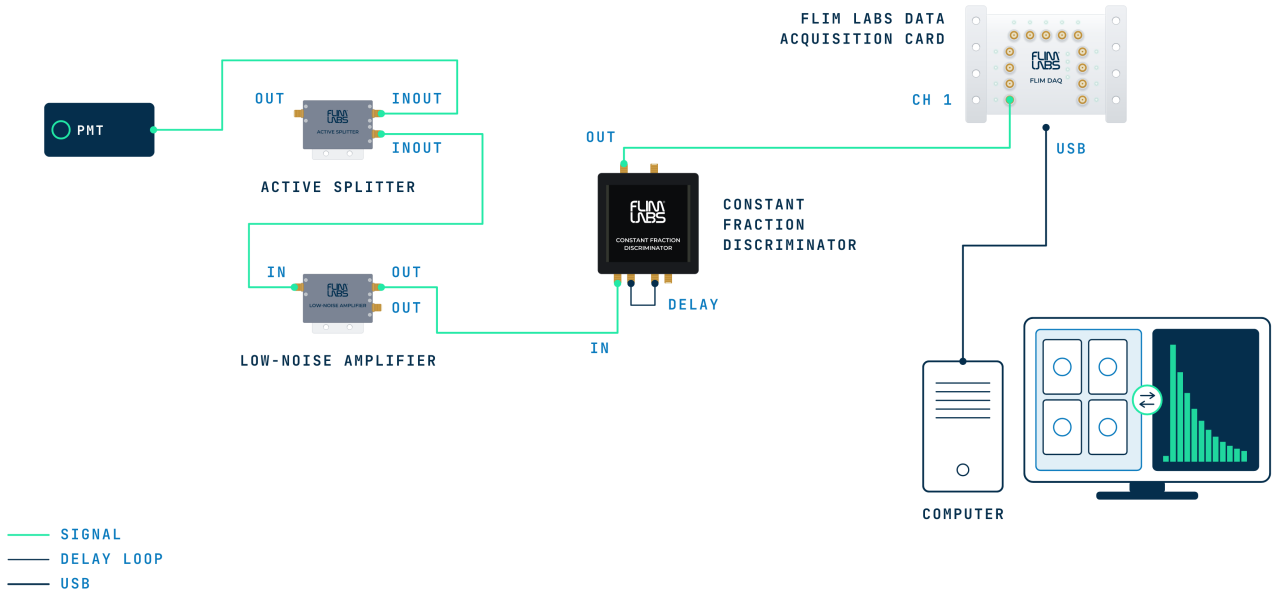


Figure 4.1: Connection example – PMT signal connection to the FLIM DAQ. The Active Splitter duplicates the PMT signal for parallel acquisition by the microscope system and the FLIM DAQ.

4.7 Initial Configuration

After completing all electrical connections, no further configuration is required.

5 Operation

This section describes the functional behavior of the Active Splitterduring use, including its operating principle, normal operating conditions, adjustable parameters, and troubleshooting guidance.

5.1 Principle of Operation

The Active Splitter distributes an input signal to multiple outputs while preserving signal integrity. The signal applied to one SMA loop-through port is forwarded unchanged to the second loop-through port and simultaneously replicated by an active stage to the dedicated output.

This enables simultaneous signal routing and duplication without significant degradation of amplitude or timing characteristics.

5.2 Normal Operation



Do not operate the device if it is damaged or shows signs of unsafe condition.

Once the power supply and signal connections have been correctly established, the device operates autonomously without further user intervention. Correct functionality depends on proper system integration and compliance with the specified signal and environmental limits.

5.3 Parameter Adjustment

The device does not require user-adjustable parameters for normal operation.

5.4 Troubleshooting

Common faults and corrective actions are summarized in Table 5.1.

Table 5.1: Overview of common problems and recommended corrective actions.

Problem	Possible Cause	Recommended Action
Power and Device Status		
No output signal	The device is not connected to a power source	Connect the device to a suitable power source according to the instructions.
The device does not power on via USB	Insufficient power supplied by the USB port	Use a USB 3.0 port or another USB power source capable of supplying 900 mA at 5 V.
The device does not power on	Defective USB cable	Replace the USB Type-C cable and verify proper connector seating.
Input Signal		
No output signal	The input signal does not meet the requirements specified in the technical data	Check the input signal with an oscilloscope or another suitable instrument and verify compliance with the input specifications.
No output signal	Input signal amplitude too low or signal type not suitable	If available, use a suitable digital signal or use a signal conversion or conditioning device.
Output signal unstable	Noise on the input signal or improper cabling	Improve signal quality, reduce cable length, and ensure proper shielding and grounding.
Output Signal		

Problem	Possible Cause	Recommended Action
No signal detected by external devices	Incorrect output connection	Verify that both SMA outputs are correctly connected to the intended external devices.
Output signal distorted	Improper impedance matching or excessive cable length	Use short 50 Ω coaxial cables and ensure proper termination of the signal path.



If any unexpected or potentially hazardous anomaly occurs during startup or operation, switch off the device immediately and disconnect it from the power source. Contact technical support before further use.

6 Maintenance and Service

This section describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the device. All maintenance activities must be performed by qualified personnel who have read and understood this manual.



Before performing any cleaning or physical maintenance operation, the device must be switched off and disconnected from the power source.

Maintenance must be carried out in compliance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.

6.1 Cleaning

The device requires periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from the exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors or internal components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas (e.g., USB ports), a soft-bristled brush or compressed air may be used.

6.2 Preventive Maintenance

The device does not require routine internal maintenance.
Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear or deformation
- Product label for legibility. See Section 1.4 for label content and handling requirements.

If any abnormal condition is detected, discontinue use and contact technical support.

6.3 Service and Repairs

The device contains no user-serviceable internal components. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the device.
- Do not attempt internal repair.
- Contact the manufacturer’s technical service for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts. Further warranty conditions are described in Section 1.6.1.



Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.

7 Decommissioning and Disposal

This section describes procedures for removing the device from service and disposing of it in accordance with applicable regulations.

7.1 Decommissioning Procedure

The device is designed for long-term laboratory use. When it reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the device from all power sources.
- Removing all signal connections.
- Ensuring that the device cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

7.2 Disposal

The device must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment. In accordance with Directive 2012/19/EU (WEEE), the device must not be disposed of with household waste. The product label includes the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment (see Section 1.4). Contact your local waste management authority or certified collection center for proper disposal procedures.

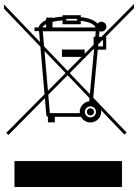


Figure 7.1: WEEE symbol